

DAVID RAJ RAMAKRISHNAN

Senior Software Engineer | Unity / C# | Technical Lead

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Professional Summary

Senior software engineer and technical lead with 10+ years in software development, including 8+ years working with Unity and C#. Experience spans interactive applications, XR, desktop and WebGL platforms, Azure integration, and AI-enabled enterprise tools. Architect and team lead for the SpeedShelf Lite WebGL pilot for Procter & Gamble at Capgemini. Strong background in reusable software architecture, UI development, memory and rendering optimization, code reviews, and technical mentoring.

Technical Skills

Development

C#, Unity3D, object-oriented programming (OOP), SOLID principles, design patterns, modular architecture, reusable components, gameplay programming

Performance

Unity Profiler, debugging, memory management, garbage collection optimization, object pooling, texture optimization, rendering performance

Engineering delivery

Git, GitHub, Sourcetree, Git LFS, Agile/Scrum, SDLC, requirements analysis, technical design, code reviews, documentation, mentoring

Unity and XR

UGUI, TextMeshPro, ScriptableObjects, editor scripting, coroutines, URP, RenderTexture, WebGL, OpenXR, XR Interaction Toolkit, Quest, LeanTouch

Cloud and integration

Azure Blob Storage, Azure App Service, Azure Container Apps, REST APIs, UnityWebRequest, Python, FastAPI, MSAL integration, OpenAI API integration

Professional Experience

Senior Consultant — Technical Lead & Solution Architect, Capgemini

06/2023 – Present

Client: Procter & Gamble

- Lead architecture and development for SpeedShelf desktop and the SpeedShelf Lite WebGL pilot, translating enterprise visualization requirements into C# components and Unity application workflows.
- Define the WebGL migration roadmap, high-level design, memory and performance strategy, and technical risks; guide implementation through architecture reviews and team mentoring.
- Develop UI workflows, custom editor tools, ScriptableObjects, planogram data import, image matching, and rendering features for retail shelf visualization.
- Integrate Azure Blob Storage and authenticated REST APIs for project files, product images, thumbnails, and cloud-based rendering workflows.
- Optimize texture loading, batching, memory use, and rendering; replace repeated product-image searches with indexed matching to reduce processing complexity.
- Integrate OpenAI APIs into shelf analytics and recommendation workflows, reducing manual review effort by approximately 80%.
- Establish coding standards and Git workflows; participate in code reviews, sprint planning, defect resolution, technical documentation, and knowledge transfer.

Professional Experience Continued

Game Engineer II — Software Engineer, Osmo (Byju's)	10/2020 – 04/2023 Bengaluru, India
• Developed reusable software frameworks and components for interactive products, supporting adoption across multiple product lines. • Improved application architecture and UI systems through collaboration with cross-functional engineering teams. • Maintained multiple Git repositories for parallel releases and contributed to bug triage, sprint planning, and Agile delivery.	
Junior Software Engineer, Tech Mahindra	08/2016 – 10/2020 India
• Developed enterprise applications integrating real-time operational data and delivered more than 20 proof-of-concept solutions for client engagements. • Collaborated with cross-functional teams to prototype solutions, validate technical approaches, and demonstrate capabilities to clients and executives. • Received recognition twice for engineering contributions and executive-level solution demonstrations.	

Selected Projects

SpeedShelf Lite — WebGL Architecture and Azure Integration,

Capgemini / Procter & Gamble | Unity, C#, WebGL, Azure Blob Storage, Python, FastAPI

- Design workflows from authenticated login and recent projects to PSA file loading, product-image retrieval, planogram visualization, and rendered output.
- Implement batched image processing, texture resizing, CPU/GPU texture handling, and product matching using normalized identifiers and folder filtering.
- Build a tiled capture and backend stitching workflow using FastAPI and ImageMagick, supporting 300 DPI output and stitched images up to approximately 25,000 × 39,000 pixels.

SpeedShelf Desktop — Retail Visualization and Rendering,

Capgemini / Procter & Gamble | Unity, C#, UGUI, URP, RenderTexture

- Develop and maintain planogram import, product-image replacement, labels, price tags, and interactive visualization workflows.
- Implement front and angled camera capture, orthographic and perspective rendering, and tiled image processing within Unity texture-size constraints.

Education

Executive MBA, Dayananda Sagar University In progress	2025 – 2027
Bachelor of Computer Applications (BCA), Periyar University	2013 – 2016

Awards And Recognition

Client appreciation for the SpeedShelf Lite pilot architecture.	01/04/2026
Customer Delight Award, Capgemini for contributions to application delivery and performance.	01/04/2025

Certification

- Unity Certified Developer